RoHS Compliant
Directive 2011/65/EU

SPECIFICATION

Customer :

Item:	CRYSTAL UNIT
Type:	NX2012SA
Nominal Frequency:	32.768kHz
Customer's Spec. No.:	---
NDK Spec. No.:	STD-MUB-1

Receipt

Charge:

Sales	Sales Dept.6 : LIU YU	Tel. 81-3-5453-6737
Engineer	5th Eng. Dept.: Hasuike	Tel. 81-4-2900-6632

Approved	H.Matsudo
Checked	---
Drawn	Y.Hasuike

Revision Record

Rev.	Rev. Date	Items	Contents	Remarks
---	20.Nov.2012	Issue	---	---

1. Customer specifications number : ---

2. NDK specification number : STD-MUB-1

3. Type : NX2012SA

4. Electrical characteristics
   - 4.1. Nominal Frequency (F_0) : 32.768 kHz
   - 4.2. Overtone Order : Fundamental
   - 4.3. Adjustment tolerance : $\pm 20 \times 10^{-6}$ Max. (at +25°C ,Not include aging)
   - 4.4. Turning Point : +25°C±5°C
   - 4.5. Temperature coefficient : -0.04×10^{-6} / °C^2 Max.
   - 4.6. Equivalent Resistance (R_R) : 80 kΩ Max.
   - 4.7. Shunt Capacitance (C_0) : 1.3 ± 0.3 pF
   - 4.8. Motional Capacitance (C_1) : 5.0 ± 1.0 fF
   - 4.9. Insulation Resistance : Terminal to terminal insulation resistance also terminal to cover insulation resistance must be 500MΩ (Min.) when DC100V ±15V is applied.

5. Measurement circuit
   - 5.1. Frequency measurement
     - ・Measuring instrument : Network Analyzer (CNA-LF made in Transat corp.)
     - ・Load capacitance (C_L) : 12.5pF
     - ・Level of drive : 0.1 μW
   - 5.2. Equivalent resistance measurement
     - ・Measuring instrument : Network Analyzer (CNA-LF made in Transat corp.)
     - ・Load capacitance (C_L) : Series
     - ・Level of drive : 0.1 μW

6. Other performances
   - 6.1. Operating Temperature range : - 40 to + 85°C
   - 6.2. Storage Temperature range : - 40 to + 85°C
   - 6.3. Maximum drive level : 0.5 μW Max.
   - 6.4. Aging (at +25 °C) : $\pm 3 \times 10^{-6}$ Max. / 1 year

7. Examination results document
   Since a performance is guaranteed, an examination results document does not submit.

8. Application drawing
   - 8.1. Dimension drawing : EXD14B-00387
   - 8.2. Taping and reel figure : EXK17B-00273
   - 8.3. Marking Structure : EXH11B-00366
   - 8.4. Reel Packing : EEK17B-00015
   - 8.5. Reliability assurance Item : EXS30B-00845

9. Notice

9.1 Order items are manufactured according to specification. As to conditions, which are not indicated in t his specification and unpredictable such as applied condition and oscillation margin, please check them beforehand.

9.2 Unless we receive request for modification within 3 weeks from the issue date of this NDK specification sheet, we will supply products according to this specification. Also, if you'd like to modify specification of order, which has been placed with delivery request within 3 weeks from the issue data of this specification sheet, we would like to discuss with you separately.

9.3 In no event shall the company be liable for any product failure resulting from an inappropriate handling or operation of the product beyond the scope of its guarantee.

9.4 Where any change to the process condition is made due to the change(s) in the production line, inform personnel of the specifications.

9.5 Should this specification data give rise to any disputes relating to any intellectual property rights or any other rights of a third person, the company shall not indemnify anyone for any damage. Their disclosure must not be construed as the grant of a license to use any of the intellectual property rights owned by the company.

9.6 If you intend to use products listed on this specification for applications that may result in loss of life or assets (controls relating to safety, medical equipment, aeronautical equipment, space equipment, etc.), please do not fail to advise us of your intention beforehand.

9.7 In the company's production process whatever amount of ozone depleting substances (ODS) as s pecified in the Montreal protocol is not used.

9.8 Information contained in this specification must not be quoted, reproduced or used for other purposes including processing either in part or in full without obtaining prior approval from the company.

10. Prohibited items

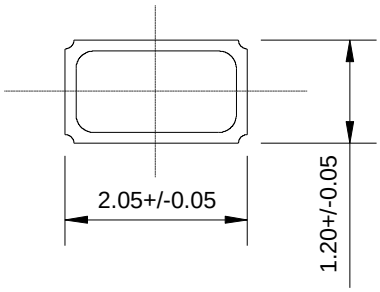
Be sure to use the product under the following conditions. Otherwise, the characteristics deterioration or destruction of the product may result.

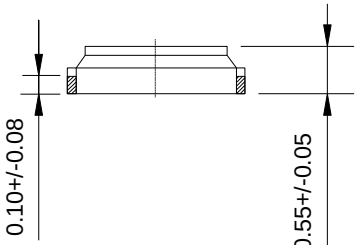
(1)Reflow soldering heat resistance

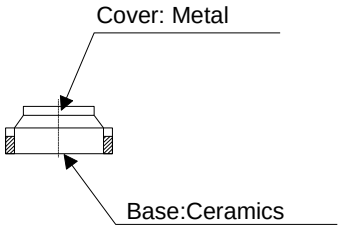
Peak temperature	: 265°C, 10 sec
Heating	: 230°C or higher, 30 sec
Preheating	: 150°C to 180°C, 120 sec
Reflow passage times	: Two times

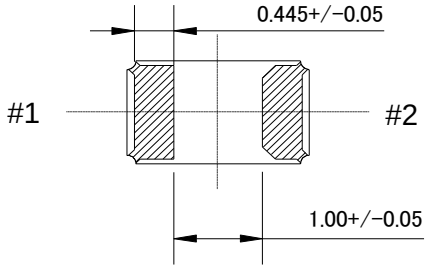
(2)Manual soldering heat resistance

Pressing a soldering iron of 400°C on the terminal electrode for four seconds (twice).

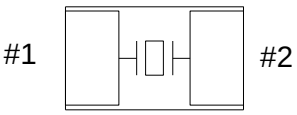




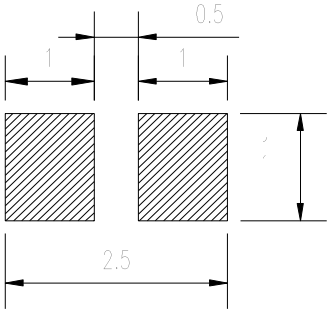




INTERNAL CONNECTION
(TOPVIEW)



Recommended soldering pattern



	Date of Revise	Charge	Approved	Reason
D	3.Jul.2012	Y.Hasuike	H.Matsudo	Addeed Castellation

	Date	Name	Third Angle Projection	Tolerance	Scale
Drawn	17.July.2007	S.Kawanishi	Dimension:mm	±0.2	10 / 1
Designed	17.July.2007	S.Kawanishi	Title	Drawing No.	Rev
Checked	17.July.2007	M.Yoshimatsu	**NX2012SA External Dimension**	**EXD14B-00387**	D
Approved	17.July.2007	K.Ono			

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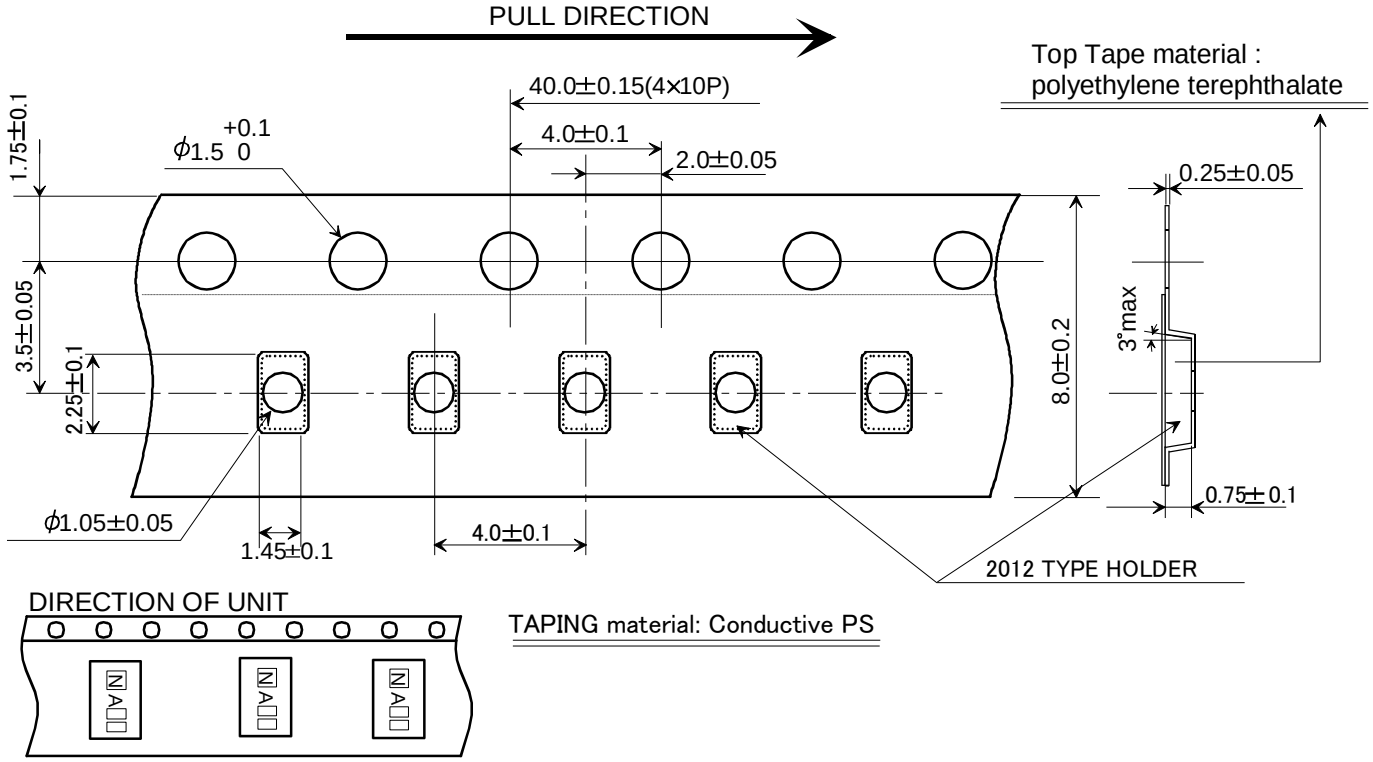

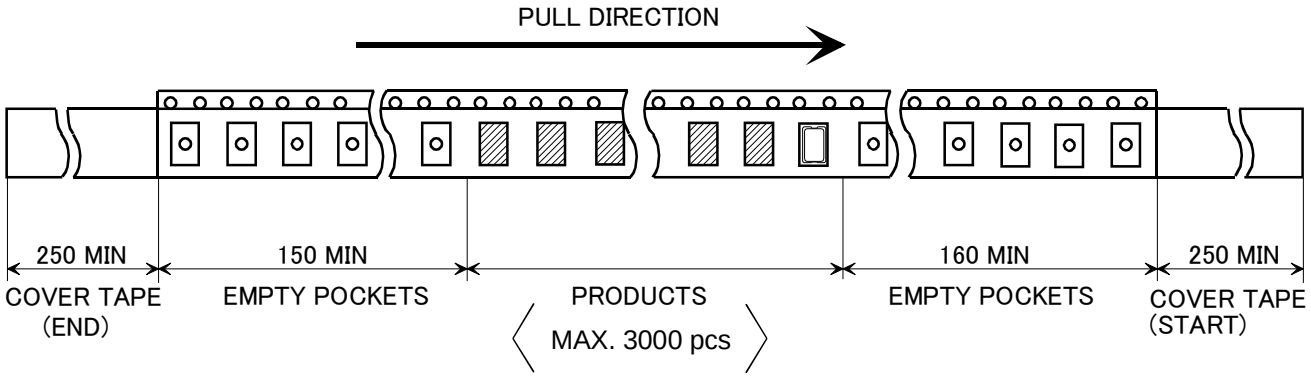

	Date of Revise	Charge	Approved	Reason
C	3 Aug.2012	Y.Hasuike	H.matsudo	Added of quantity

	Date	Name	Third Angle Projection	Tolerance	Scale
Drawn	31.Jul.2007	K.Oguri	Dimension:mm		/
Designed	31.Jul.2007	S. Kawanishi	Title	Drawing No.	Rev.
Checked	----------	----------	**2012 TYPE Taping and Reel Spec.**	**EXK17B-00273 1/2**	C
Approved	31.Jul.2007	K. Ono			

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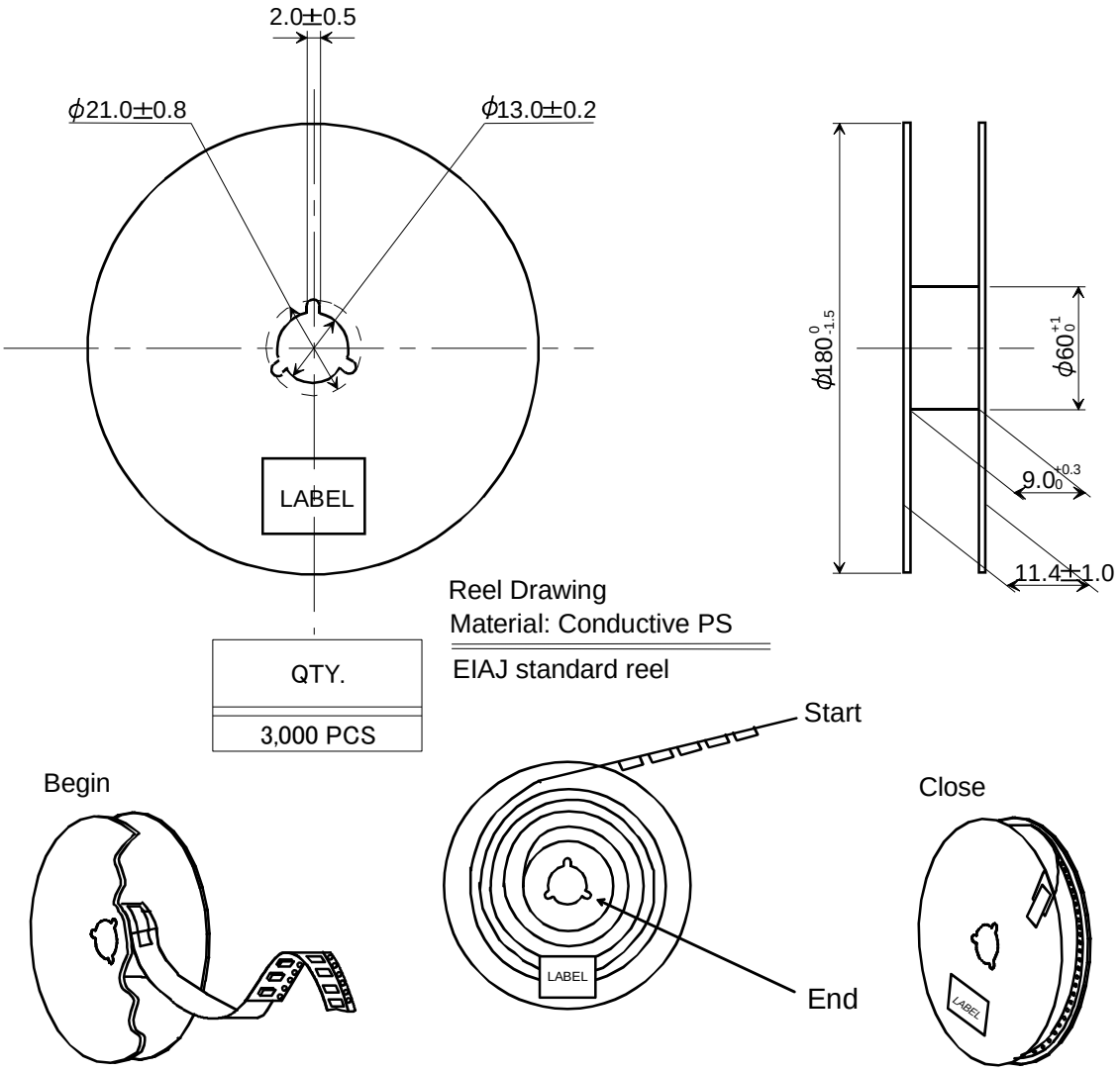

	Date of Revise	Charge	Approved	Reason
C	3 Aug.2012	Y.Hasuike	H.matsudo	Added of quantity

	Date	Name	Third Angle Projection	Tolerance	Scale
Drawn	31.Jul.2007	K.Oguri	Dimension:mm		/
Designed	31.Jul.2007	S. Kawanishi	Title	Drawing No.	Rev.
Checked	----------	----------	**2012 TYPE Taping and Reel Spec.**	**EXK17B-00273 2/2**	C
Approved	31.Jul.2007	K. Ono			

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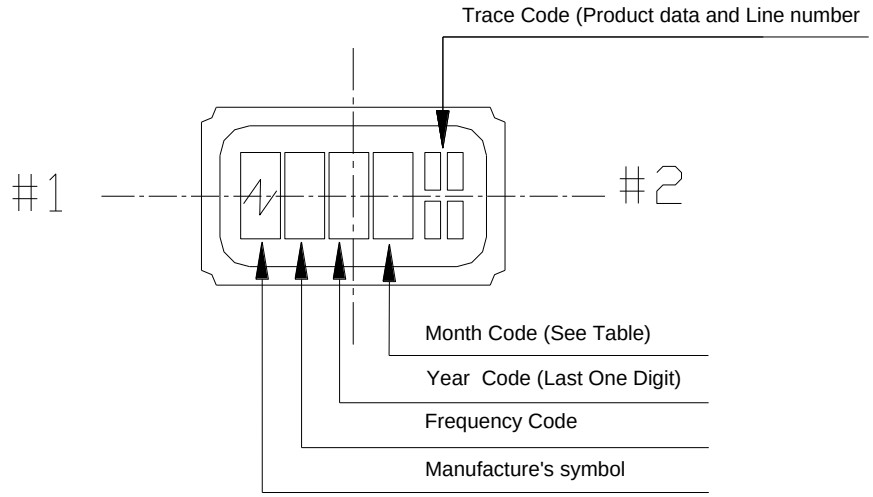
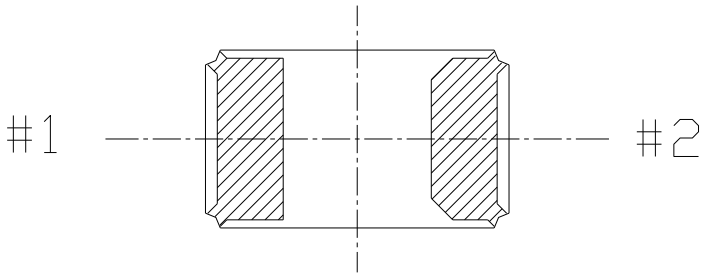
Form M-1





NOTE

1. Month Code

Month	1 Jan.	2 Feb.	3 Mar.	4 Apr.	5 May	6 June	7 July	8 Aug.	9 Sep.	10 Oct.	11 Nov.	12 Dec.
Month Code	1	2	3	4	5	6	7	8	9	X	Y	Z

2. Frequency Code

   A : 32.768kHz

3. Marking Method

   Marking Method is Laser Trimming.

	Date of Revise	Charge	Approved	Reason
B	9.June.2010	S.Kawanishi	M.Umeki	To change the direction of crystal unit

	Date	Name	Third Angle Projection	Tolerance	Scale
Drawn	20.July.2007	S.Kawanishi	Dimension:mm		/
Designed	20.July.2007	S.Kawanishi	Title: **NX2012SA Marking Drawing**	Drawing No.: **EXH11B-00366**	Rev.: B
Checked	20.July.2007	M.Yoshimatsu			
Approved	20.July.2007	K.Ono			

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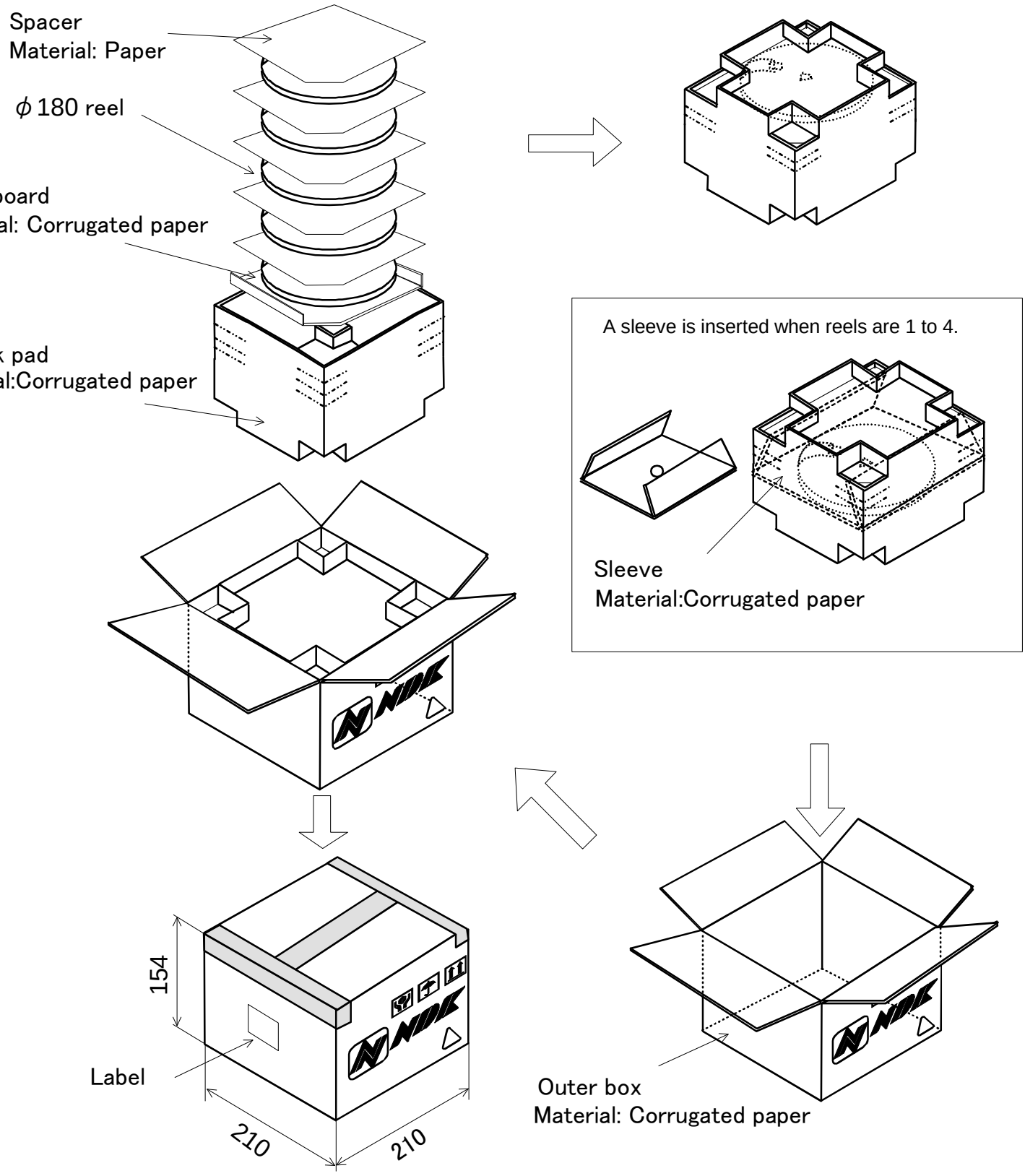

	Date of Revise	Charge	Approved	Reason
C	4 Jul. 2012	H.Ohkubo	K.Oguri	Addition of condition when reels are 1 to 4.

	Date	Name	Third Angle Projection	Tolerance	Scale
Drawn	26 Feb. 2010	H. Ohkubo	Dimension:mm	---------	---------
Designed	26 Feb. 2010	K.Oguri	Title	Drawing No.	Rev.
Checked	26 Feb. 2010	K.Oguri	**180 dia. Reel package**	**EEK17B-00015**	C
Approved	26 Feb. 2010	J. Nakamura			

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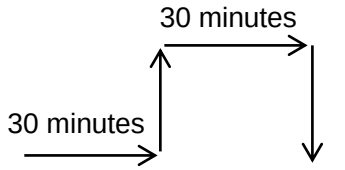
Reliability assurance item

No.	Test Item	Test Methods	Specification Code
1	HEAT RESISTANCE	at +85 °C for 1000 hours.	a
2	COLD RESISTANCE	at –40 °C for 1000 hours.	a
3	HUMIDITY	at +85 °C with 80 to 85 % RH for 1000 hours.	a
4	THERMAL SHOCK	Temperature cycle as shown in (Fig.1) for 1000 cycle. +85 °C +/- 3 °C -40 °C +/- 3 °C  ONE CYCLE (Fig.1)	a
5	VIBRATION	Frequency Range : 10 to 2000Hz Amplitude or Acceleration : 1.52 mm or 20 G 1 cycle : 20 minutes Test time : Three mutually perpendicular axes each 12 times.	a
6	SHOCK 1	Shock : 3000 Gs 0.3 msec. Test time : Six mutually perpendicular axes each 1 times.	a
7	SHOCK 2	Shock : Device are put on the weight of 140 g and dropped on concrete board. Height : 1.5 m Drop times : Three mutually perpendicular axes each 10 times.	b
8	SOLDERABILITY	Residual heat temperature 150 °C Residual heat time 60 to 120 sec Peak temperature 240°C (more than 215 °C 10 to 30 sec)	c
9	REFLOW RESISTANCE	Temperature cycle as shown in (Fig2.) for 3 cycle.	a

Specification code	Specification
a	dF/F ≤ +/- 10ppm dCI ≤ +/- 20 kohm
b	dF/F ≤ +/- 20ppm dCI ≤ +/- 20 kohm
c	The electrodes shall acquire a new solder coat over at least 90 % of immersed area.

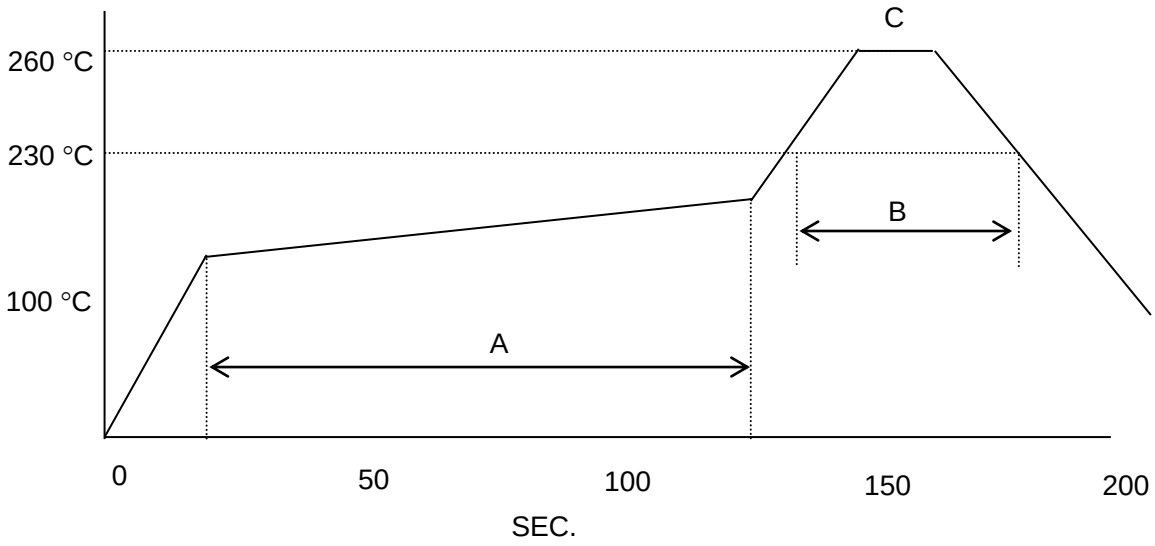


Fig.2 REFLOW

A: 150 to 180 °C (60 to 120 sec.)
B: 230 °C min. (30 sec. max.)
C: PEAK-TEMP. 260 °C +/- 5 °C (10sec. max.)